A Full DR11 Strong-Lens Search in the DESI Legacy Surveys: ClaudeNet `v3blend8` Finding with LensJudge v3 HSC Cross-Validation

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Abstract

We deploy the contaminant-aware ClaudeNet `v3blend8` ensemble and the LensJudge v3 vetting cascade on the final DESI Legacy Surveys data release (DR11). The **DR11-south** (DECam, native *griz*) sweep scores all 5.38×10^7 galaxies with the five-member v2-lean stage and re-ranks the survivors with the eight-member `v3blend8` ensemble; a DR11-native recalibration is required because the deeper DR11 reductions have fatter random-galaxy score tails than DR9, tightening the matched-FPR thresholds (known-lens grade-A recall at the 10^{-4} operating point falls 62% → 41% relative to DR10 — a genuine DR9→DR11 domain shift, not a threshold artifact, so the DR11 survivor list under-recovers grade-A lenses and is not complete). Group-conformal selection remains *power-limited* at $m \approx 5.4 \times 10^7$ (zero certified at $\alpha \leq 0.25$ with an 8M-negative calibration), so the candidate list is the `v3blend8`-ranked top-N, as in the prior campaigns. The decisive step is LensJudge v3's new **HSC PDR3 tier-2**: vetting the top-500 new candidates as a cascade (cheap detection triage → agentic mimic adjudication → high-resolution escalation, \$40 total) moves **26** HSC-covered candidates from DESI-ambiguous ($p_{\rm lens} \sim 0.03$–0.6) to **24 HSC tier-2 grade-A/B** under LensJudge's automated grader ($p_{\rm lens}$ up to 0.97; a single automated grader, not a human committee or spectroscopy). An independent SuGOHI cross-match calibrates the engine: `v3blend8` recovers **79** SuGOHI HSC committee lenses into the survivors at 2.7× the survivor-median score, and **15 of the 24** graded A/B are SuGOHI lenses (a sensitivity check — the automated tier-2 grader's precision on the genuinely-new candidates is not established here). The remaining **9** (8 distinct systems) are new to the DESI lens-finder catalogs and SuGOHI; a post-hoc literature crosscheck (Huang+2020, omitted from the campaign crossmatch, + NED/SIMBAD; App. A) then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also SOGRAS), leaving **3** genuinely new — automated-grader candidates pending follow-up. The honest frame is unchanged from the v3 program: the model recovers known lenses and ranks new candidates the higher-resolution HSC grader rates highly, but *no net-new lens is established from DECaLS pixels alone* (the new candidates depend on HSC resolution and remain unconfirmed), and net-new discovery beyond existing high-resolution catalogs is resolution-bounded — the value of the *resolution lever* (now HSC over ∼1,300 deg^2, far wider than Euclid Q1) is to *convert* overlap candidates, not to manufacture discoveries. The **DR11-north** (BASS/MzLS) arm — which needs instrument-native retraining (v3 is DECam-specific) — is fully staged but deferred by a 7-day Perlmutter maintenance.

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1 Premise

The ClaudeNet v3 program concluded that for DECaLS-resolution lens finding the binding constraint is *lens-vs-mimic separation* and *vetting resolution*, not architecture: `v3blend8` (the SHIP-gated eight-member ensemble: `effnet_B`, `zoobot_N`, and the v2-hard + v3-mimic-blend versions of `effnet_S2`, `effnet_B3`, `resnet46_C`) rejects common contaminants far better than v2 but is statistically tied with the originals at the hardest real-vs-mimic frontier, which is resolution-bounded (Inchausti Reyes et al., 2025). The productive mode is therefore *DESI pre-screen → high-resolution confirmation in the overlap*. This campaign operationalizes that on the final release (DR11) and exploits the new lever in LensJudge v3: an **HSC PDR3** tier-2 grader (Aihara et al., 2022) ($0.168''$/px, $\sim 0.6''$ seeing) whose footprint (HSC-SSP Wide, $\sim$1,300 deg^2) is $\sim 20\times$ the Euclid Q1 area used previously, vastly widening the confirmable overlap.

2 The DR11-south sweep

DR11-south is the deepest DECam reduction to date: 1,600 sweep files (1.7 TB) yielding a parent of **53,809,040** galaxies after the published Inchausti cuts (`TYPE`$\in${SER,EXP,DEV,REX}, NOBS$_{grz} \geq 3$, $m_z < 20$), 51.4M with native i. We shard into 32 footprint-pure, brick-disjoint parts, extract 101-px grz cutouts (100% success, 6.0 TB), and score every galaxy with the five v2-lean members.

DR11-native recalibration (required). The DR9-fit per-member 10^{-4}-FPR thresholds, when transferred to DR11 score-space, are *too tight*: DR11's deeper imaging produces a fatter high-score tail of random galaxies (e.g. the `resnet46_C` isotonic score saturates, its 10^{-4} threshold moving $0.81 \to 1.00$, effectively reducing the union to four members). We recalibrate from the scored population itself (a 8 M-galaxy random NegEval), yielding **95,104** survivors. Re-measuring recall of the published Inchausti 811 against these survivors gives grade-A 41% (37/90), versus 62% for the DR10-recalibrated DR10 sweep — a real DR9→DR11 model domain shift (the DR9-trained members under-score a fraction of DR11 grade-A lenses, *and* the random tail is fatter), not a threshold artifact (the denominator includes north and parent-cut lenses, so 41% is a conservative floor).

Selection. Scoring the survivors with all eight `v3blend8` members and ranking by the mean calibrated score, the top-300 contain **102 known** lenses (recovered into the top of the list — a consistency check) and **198 new** candidates (top-300 `v3blend8` score 0.812–0.939). The discovery set is the **top-500 new** (`v3blend8` score 0.734–0.933); the broader survivor pool has 1,071 survivors above `v3blend8` 0.7 (726 new + 345 already-known).

Certified-FDR (power-limited, reported honestly). Group-conformal selection (Jin and Candès, 2023) with the 8 M NegEval certifies **zero** candidates at $\alpha = 0.05/0.10/0.25$: the per-group p-value floor $1/(n_{\text{cal}}+1) \approx 2.5 \times 10^{-7}$ is $\sim 50\times$ short of what full-m Benjamini-Hochberg needs at $m = 5.38 \times 10^7$ to select even one. This is the same power-floor diagnosed at DR9/DR10 scale — rigorous full-population FDR needs an infeasibly large calibration set — so the candidate list is the `v3blend8`-ranked top-N, not an FDR-certified set. The arithmetic self-test (14/14) and the byte-level scoring-path check pass.

Table 1: DR11-south cascade outcome (500 new candidates, `pass_frac`=0.7).

quantity	value	note
graded (stage-1 triage)	500	\$0.012 each
escalated (agentic stage-2)	350	v2 mimic rubric
reached HSC tier-2	26	HSC-SSP Wide coverage
grade A / B / C / D	27 / 76 / 125 / 272	final (DESI- or HSC-grade)
HSC tier-2 grade-A/B	**24** (19 A, 5 B)	automated; 8 more A are DESI-only
total cost	\$40.38	

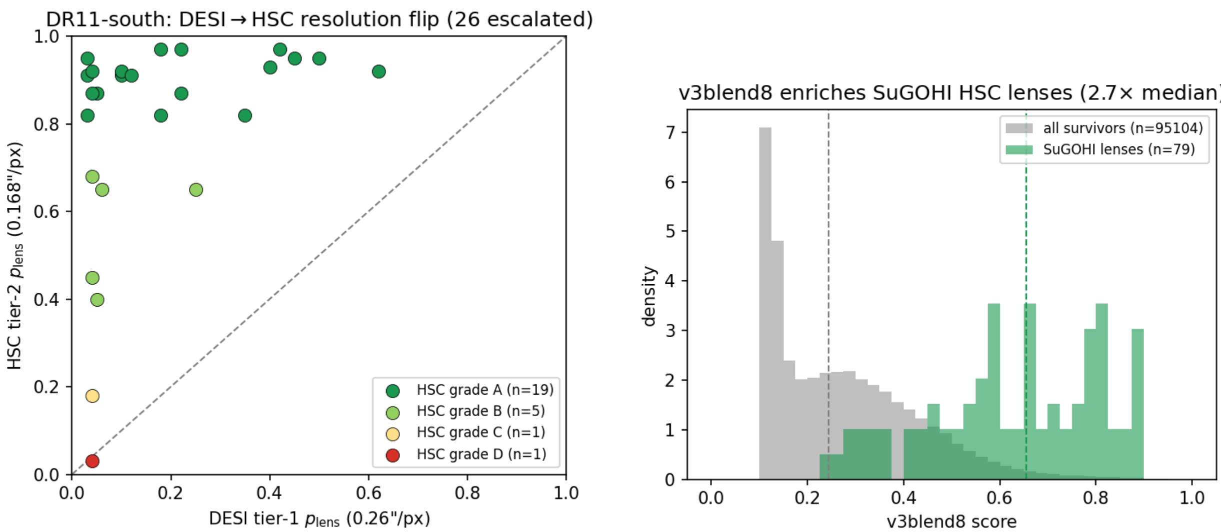

Figure 1: Left: the DESI→HSC resolution flip — the 26 HSC-covered candidates move from DESI-ambiguous (tier-1 $p_{\rm lens}$) to the HSC tier-2 grade (automated grader), 24 landing A/B above the $y\!=\!x$ line. Right: `v3blend8` enriches the independent SuGOHI HSC lenses to 2.7× the survivor-median score (median lines dashed).

3 LensJudge v3 cascade vetting + HSC cross-validation

The DESI viewer exposes no public DR11 layer (and is unreachable from our host), so we stage the 500 candidates' *grz* cubes from Perlmutter as on-disk FITS, ensuring LensJudge grades the exact pixels the CNN scored. We then run the B7 deploy cascade: a loop-free `direct` detection triage on all 500 (∼\$0.012 each), escalating the top 70% by rank to the agentic v2 mimic-adjudication grader, which itself escalates to **HSC PDR3 tier-2** wherever HSC covers the position. Total cost **\$40.4** (campaign total \$43 of a \$250 budget).

The resolution lever converts. Every tier-2 grade-A/B candidate shows a large DESI→HSC flip: median tier-1 $p_{\rm lens}\!\approx\!0.1$ rising to tier-2 0.82–0.97 (e.g. 0.03 → 0.95, 0.04 → 0.92). At DECaLS 1″ seeing these are ambiguous (lrg+companion ≈ lens); at HSC 0.6″ the arc/ring morphology resolves — the flip is the grader's response to resolution, not an observational confirmation.

Table 2: The 9 HSC tier-2 grade-A/B candidate detections (8 distinct systems). A literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) finds **5 systems are previously-published** (*known*: H20 = Huang+2020; one also SOGRAS) and **3** genuinely new. p_1, p_2 = LensJudge tier-1 (DESI) / tier-2 (HSC) lens probability.

name	RA	Dec	grade	$p_1 \to p_2$	`v3blend8`	known
s_340513_3688	16.332	+1.749	A	0.22 → 0.97	0.77	Huang+2020
s_327526_9059	9.722	−0.409	A	0.04 → 0.92	0.74	**new**
s_351303_461	194.534	+3.536	A	0.62 → 0.92	0.89	Huang+2020
s_345385_1515†	154.312	+2.489	A	0.10 → 0.92	0.81	Huang+2020
s_340971_2860	130.857	+1.784	A	0.12 → 0.91	0.76	**new**
s_345385_1514†	154.312	+2.488	A	0.05 → 0.87	0.86	Huang+2020
s_326089_1785	10.288	−0.730	A	0.03 → 0.82	0.83	H20+SOGRAS
s_325163_2981	138.848	−0.923	A	0.18 → 0.82	0.74	**new**
s_318043_1037	158.789	−2.304	B	0.04 → 0.45	0.86	Huang+2020

† s_345385_1514/1515 are adjacent (∼1″) detections of the same system, so the 9 detections are **8 distinct candidate systems** (8 A-detections / 7 distinct grade-A + 1 grade-B).

Independent SuGOHI validation. Against the SuGOHI/HOLISMOKES HSC committee lens catalog — an *independent* sample with expert grades and spectroscopic redshifts (Sonnenfeld et al., 2018) — `v3blend8` recovers **79** lenses into the 95,104 survivors at a median `v3blend8` score 2.7× the survivor median (27 in the top-500), and **15 of the 24** tier-2 grade-A/B candidates are SuGOHI lenses. These 15 are a *sensitivity* check — the automated HSC grader recovers known committee lenses as grade-A/B — but they do not establish its *precision* on the genuinely-new candidates: no negative control was run through tier-2 and the certified-FDR is power-limited (§2), so the false-positive rate of the new set is unquantified.

The candidate catalog. Of the 24 tier-2 grade-A/B candidates, **0** are in the DESI lens-finder catalogs (by construction — the 500 were the new set after removing the 102 known), 15 are SuGOHI cross-matches, and **9** (8 distinct systems) are new to the DESI finders and SuGOHI (Table 2). A post-hoc literature crosscheck — Huang+2020 (omitted from the campaign's lineage crossmatch) plus a NED/SIMBAD cone search (App. A) — then shows **five of these eight systems are previously-published lenses** (mostly Huang+2020; one also in the SOAR Gravitational Arc Survey), leaving **3** genuinely new (`s_327526_9059`, `s_340971_2860`, `s_325163_2981`); the recoveries are an external validation of the finder. The honest caveats: LensJudge's HSC grading is automated (a single Claude grader, not a human committee, no spectroscopy), and "new" here means new to the DESI finders, SuGOHI, *and* the NED/SIMBAD literature. They are candidates warranting human-expert and spectroscopic follow-up, not confirmed lenses.

Active-learning loop. The cascade's agent-confirmed mimics (grade-D + named contaminant) export to **291** hard negatives in the ClaudeNet mimic-bank schema — `lrg_companion` 243-dominant, confirming that the high-`v3blend8` new frontier is lrg+companion-dominated at DECaLS resolution (exactly the population the HSC tier-2 rejects). Folding these into the bank and a model re-iteration are Perlmutter GPU work, deferred (below).

4 DR11-north: staged, maintenance-deferred

v3 is DECam-specific — it systematically under-scores BASS/MzLS north imaging — so the north arm requires instrument-native retraining. We built the full north parent (1.16×10^7 galaxies, *grz* only, no i), an 8-part manifest, and extracted **523** north-native training positives (297 BASS/MzLS curated + 226 non-leaked catalog lenses; 86 clean held-out; 33 v1 leaks tracked); the retrainer is patched to fine-tune the three swappable members from their `_b50` checkpoints on a north-native mimic blend. A full-system Perlmutter maintenance (2026-06-17 to 06-24) began mid-campaign before the north extraction could be scheduled, so the north sweep + retrain + validation gates are **deferred to post-maintenance**; all inputs are staged for a clean resume.

5 Conclusion

On the final DESI Legacy Surveys release, the contaminant-aware ClaudeNet finder plus the LensJudge v3 HSC cascade is a working DESI×HSC cross-validation engine on the DECam-south footprint: it scores 5.4×10^7 galaxies, recalibrates honestly to the DR11 domain shift (with a real $\sim$34% relative grade-A recall loss vs DR10 — the survivor list is incomplete), and — through the new HSC tier-2 over a 20×-wider footprint than Euclid Q1 — moves 24 DESI-ambiguous candidates to HSC tier-2 grade-A/B under LensJudge's automated grader, of which 15 are SuGOHI committee lenses (a sensitivity check) and 9 (8 systems) are new to the DESI finders and SuGOHI — though a literature crosscheck (Huang+2020 + NED/SIMBAD; App. A) then shows 5 of those 8 systems are previously-published lenses, leaving **3** genuinely new. The certified-FDR is power-limited (zero) at full-population scale, the tier-2 grader is automated (no human committee or spectroscopy, precision on the new set unquantified), and *no net-new lens is established from DECaLS pixels alone* — the new candidate detections depend on HSC resolution and remain unconfirmed pending follow-up. This is the same honest ceiling the v3 program established — the resolution lever *converts* overlap candidates rather than manufacturing discoveries — now measured on DR11 with a much wider confirmable overlap. The north footprint, requiring instrument-native retraining, is staged for completion after the Perlmutter maintenance window.

References

Hiroaki Aihara et al. Third data release of the hyper suprime-cam subaru strategic program. *Publications of the Astronomical Society of Japan*, 74(2):247–272, 2022.

J. Inchausti Reyes et al. Strong lens discovery in DR10 with two deep-learning architectures. *arXiv:2508.20087*, 2025.

Ying Jin and Emmanuel J. Candès. Selection by prediction with conformal p-values. *JMLR*, 24, 2023.

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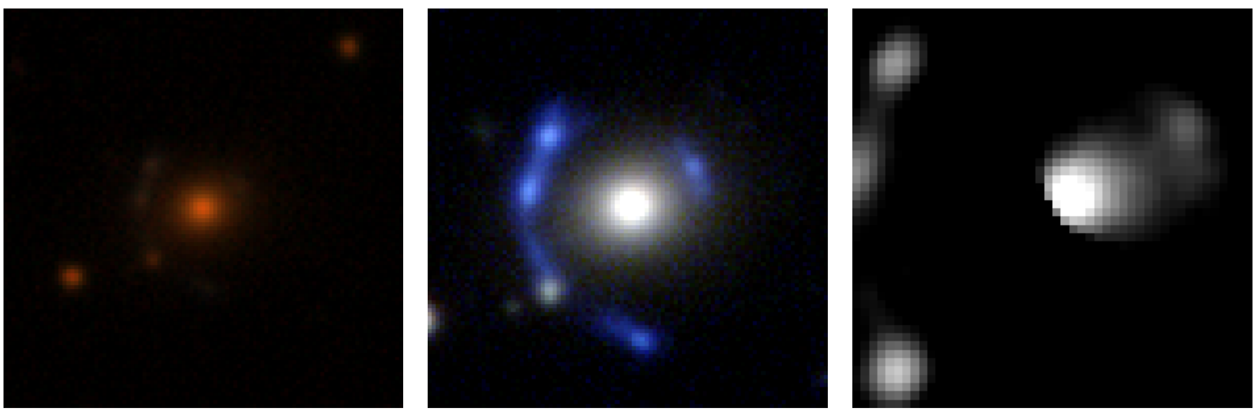
A The new candidate systems at DESI and HSC resolution

The figures below show the **8 distinct candidate systems** (the 9 detections of Table 2; the s_345385 pair is one system). A post-hoc literature crosscheck (below) finds **5 of the 8 are previously-published lenses** (Huang+2020; one also SOGRAS) and **3** are genuinely new. Each is rendered as: the ClaudeNet-seen DESI *grz* cutout (0.262″/px, Lupton RGB — what the finder scored, ambiguous at DECaLS resolution); the HSC PDR3 *grizy* cutout (0.168″/px — the same view LensJudge's tier-2 grader inspected); and the HSC lens-light-subtracted view. The DESI→HSC contrast is the campaign's mechanism made visual: the finder ranks these high from ambiguous DECaLS pixels, and the resolution lever reveals the arc/ring morphology (clearest in s_340513_3688; fainter and correspondingly less certain in others). These are *automated-grader candidates, not confirmed lenses* — they warrant human-expert and spectroscopic follow-up. *Literature crosscheck (added post-vetting).* The campaign's lineage crossmatch omitted the Huang+2020 DECaLS catalogue and did not query the wider literature. A full check — Huang+2020 plus a NED/SIMBAD cone search — finds that **five of these eight systems are previously-published lenses** (flagged *Known* below; mostly Huang+2020, one also in the SOAR Gravitational Arc Survey). Only s_327526_9059, s_340971_2860, and s_325163_2981 are genuinely new (absent from the lineage *and* the literature). The recoveries are an external validation of the v3+cascade finder; the net-new count here is **3**, not 8.

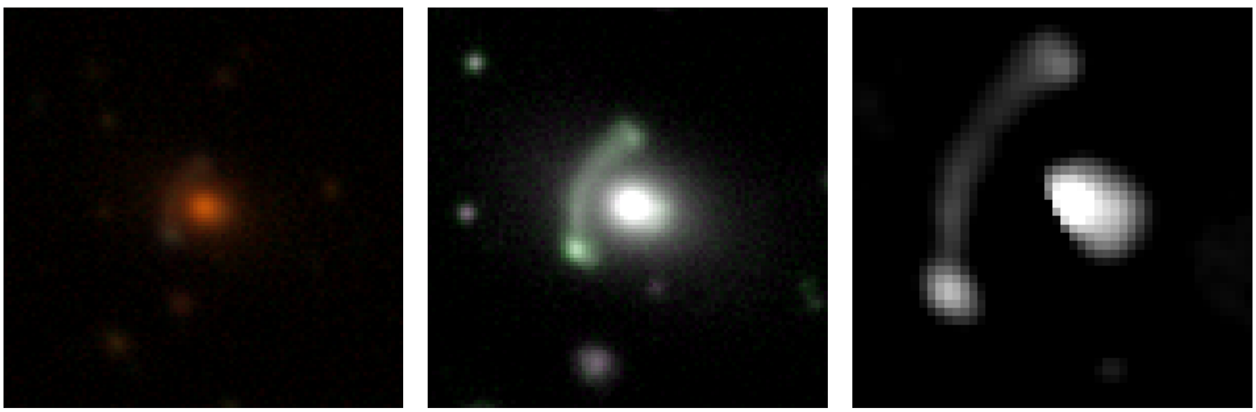


DESI *grz* (0.262″/px) | HSC *grizy* (0.168″/px) | HSC lens-light subtracted

Figure 2: s_340513_3688 (RA 16.3319, Dec +1.7490) — HSC tier-2 grade **A**, p_{lens} 0.22→0.97, v3blend8 0.77. *Known*: Huang+2020 DESI-016.3319+01.7490 (grade A), 0.1″.

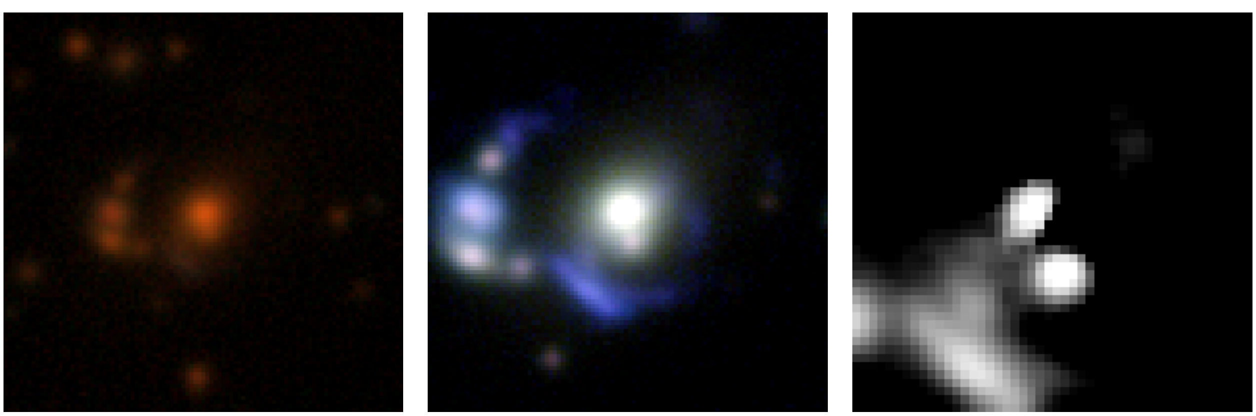
DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 3: `s_327526_9059` (RA 9.7222, Dec -0.4093) — HSC tier-2 grade **A**, $p_{\rm lens}$ 0.04→0.92, `v3blend8` 0.74. *New*: absent from the lineage and from NED/SIMBAD.



DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 4: `s_351303_461` (RA 194.5345, Dec +3.5358) — HSC tier-2 grade **A**, $p_{\rm lens}$ 0.62→0.92, `v3blend8` 0.89. *Known*: Huang+2020 `DESI-194.5344+03.5358` (grade A), 0.2″.

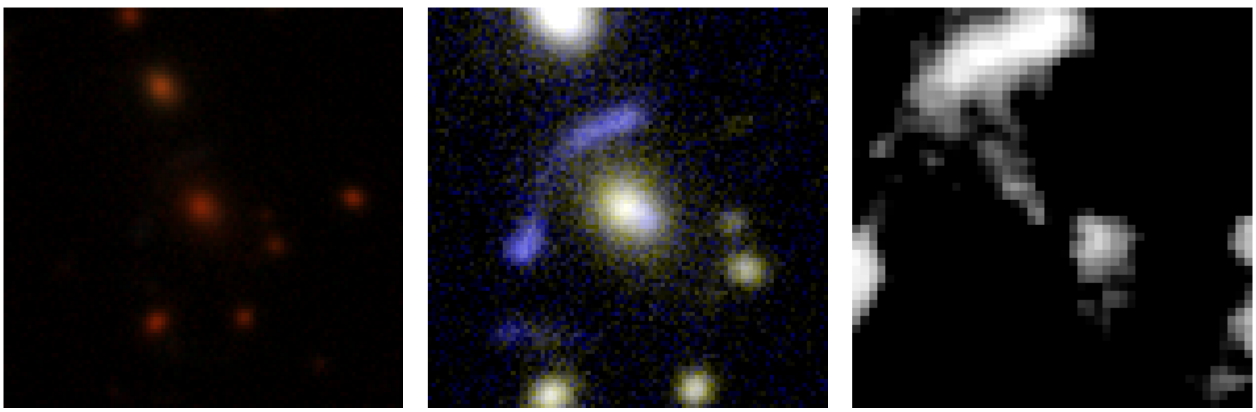


DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 5: `s_345385_1515` (RA 154.3116, Dec +2.4885) — HSC tier-2 grade **A**, $p_{\rm lens}$ 0.10→0.92, `v3blend8` 0.81. *Known*: Huang+2020 `DESI-154.3116+02.4885` (grade C), 0.1″. (= 8th detection s_345385_1514, the adjacent ∼1″ pair, omitted)

DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 6: `s_340971_2860` (RA 130.8571, Dec +1.7840) — HSC tier-2 grade **A**, p_{lens} 0.12→0.91, `v3blend8` 0.76. *New*: absent from the lineage and from NED/SIMBAD.

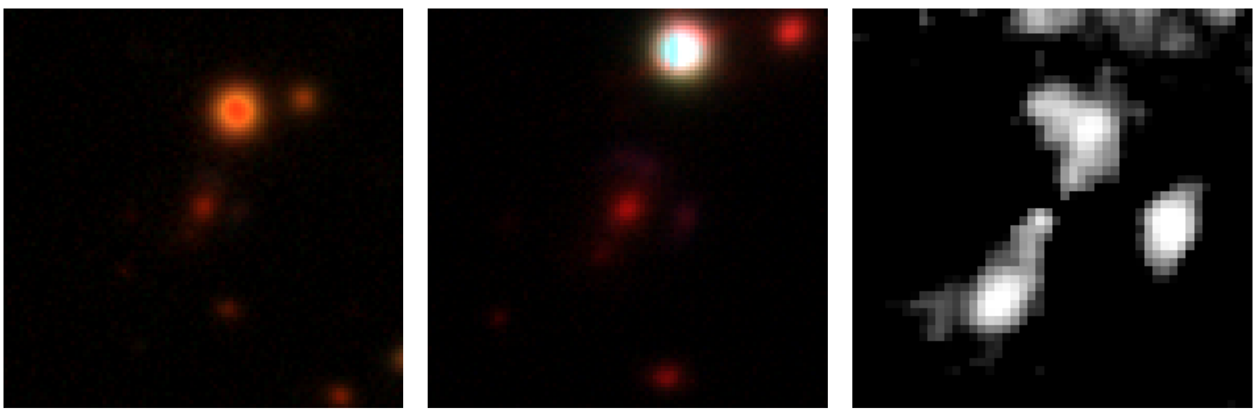


DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 7: `s_326089_1785` (RA 10.2876, Dec -0.7303) — HSC tier-2 grade **A**, p_{lens} 0.03→0.82, `v3blend8` 0.83. *Known*: Huang+2020 `DESI-010.2876-00.7303` (grade B), also SOGRAS J0041−0043, 0.1″.

DESI *grz* (0.262″/px) HSC *grizy* (0.168″/px) HSC lens-light subtracted

Figure 8: `s_325163_2981` (RA 138.8482, Dec -0.9226) — HSC tier-2 grade **A**, p_{lens} 0.18→0.82, `v3blend8` 0.74. *New*: absent from the lineage and from NED/SIMBAD.



DESI *grz* (0.262″/px) | HSC *grizy* (0.168″/px) | HSC lens-light subtracted

Figure 9: `s_318043_1037` (RA 158.7893, Dec -2.3037) — HSC tier-2 grade **B**, $p_{\rm lens}$ 0.04→0.45, `v3blend8` 0.86. *Known*: Huang+2020 `DESI-158.7893-02.3037` (grade B), 0.1″.

B DESI-resolution candidates without high-resolution coverage

Only ∼7% of the DR11-south candidates fall in HSC-SSP (§3); the rest cannot be validated beyond DECaLS resolution with current high-res data. Table 3 lists all **78** candidates graded A/B at DECaLS 1″ by the LensJudge cascade that are *not* in HSC-SSP *or* Euclid Q1 (8 A, 70 B), ranked by tier-1 p_{lens}, with DESI *grz* cutouts in Figs. 10–11. A literature crosscheck (`crosscheck_candidates.py`; Huang/Storfer/Inchausti + NED/SIMBAD) finds **53 of the 78 are previously-published lenses** — 38 from the Dark Energy Survey lens searches, 8 Huang+2020, 6 AGEL, 1 CASSOWARY (the *known* column of Table 3) — because this DECaLS-south footprint heavily overlaps DES, which the campaign crossmatch omitted; the **25 genuinely new** are all grade B (the 8 grade-A are all known lenses). At 1″ a genuine lens and an lrg+companion mimic are not separable (the resolution ceiling of §3), so these are *candidates*, an unknown and likely large fraction of which are mimics — the contact sheets show both clear arc/ring morphologies and ambiguous lrg+companion configurations. They are listed in full for completeness and as a follow-up target set, awaiting Euclid DR1 / Rubin / Roman or targeted high-resolution and spectroscopic follow-up.

Table 3: The 78 DESI-resolution grade-A/B candidates (LensJudge `direct/escalate` grade at DECaLS 0.262″) with *no* HSC-SSP or Euclid Q1 coverage, ranked by tier-1 p_{lens}. A literature crosscheck (`crosscheck_candidates.py`: the Huang/Storfer/Inchausti catalogues + a NED/SIMBAD cone search) finds **53 are previously-published lenses** — 38 DES, 8 Huang+2020, 6 AGEL, 1 CASSOWARY (*known* column: survey code; per-object matches in `data/dr11s_resolution_xmatch.csv`) — and **25 are genuinely new** (—), all grade B. The high known fraction is because this DECaLS-south footprint heavily overlaps the Dark Energy Survey lens searches, which the campaign crossmatch omitted. The 25 new ones cannot be validated beyond DECaLS resolution; at 1″ a genuine lens and an lrg+companion mimic are not separable, so an unknown fraction are mimics. Images in Figs. 10–11.

#	name	RA	Dec	grade	p_{lens}	`v3blend8`	known
1	s_310364_6388	38.2078	-3.3906	A	0.90	0.82	H20
2	s_137502_1047	26.4450	-35.6910	A	0.90	0.80	DES
3	s_181942_4122	61.6016	-26.7733	A	0.87	0.86	DES
4	s_441355_1188	177.1381	+19.5009	A	0.87	0.83	CSWA
5	s_98012_247	56.1462	-44.7956	A	0.84	0.86	DES
6	s_124958_7349	58.1768	-38.4292	A	0.81	0.89	DES
7	s_191004_9637	56.9356	-24.9088	A	0.80	0.88	DES
8	s_80200_6872	15.4919	-49.2940	A	0.78	0.88	DES
9	s_230539_4159	222.0715	-17.8489	B	0.72	0.84	—
10	s_180593_5604	44.0974	-27.1218	B	0.63	0.88	DES
11	s_292418_3907	216.8280	-6.7541	B	0.62	0.89	H20
12	s_232538_8716	25.2755	-17.2232	B	0.62	0.85	H20
13	s_234689_542	227.0427	-16.8771	B	0.62	0.80	—
14	s_42996_3467	53.7444	-60.5805	B	0.60	0.90	DES
15	s_205319_6061	359.9772	-22.4596	B	0.60	0.88	—
16	s_188493_9685	83.4556	-25.6151	B	0.60	0.88	DES
17	s_38835_314	56.6543	-61.9792	B	0.60	0.84	DES
18	s_218895_5088	32.7525	-19.6362	B	0.60	0.81	—
19	s_240846_8850	30.2832	-15.8547	B	0.60	0.80	H20

continued. . .

Table 3 continued

#	name	RA	Dec	grade	p_{lens}	v3blend8	known
20	s_111542_5062	4.8181	-41.6141	B	0.60	0.77	DES
21	s_290988_3355	217.1429	-7.0963	B	0.58	0.75	H20
22	s_194300_2796	242.6052	-24.4155	B	0.58	0.74	—
23	s_50907_11900	303.5808	-57.9504	B	0.58	0.74	DES
24	s_34114_11363	353.7468	-64.0687	B	0.55	0.90	DES
25	s_61626_1035	65.1782	-54.3770	B	0.55	0.87	DES
26	s_159172_8702	37.5699	-31.3669	B	0.55	0.82	DES
27	s_197474_9756	30.7668	-23.6341	B	0.55	0.81	DES
28	s_23764_4161	132.6603	-68.1876	B	0.55	0.79	—
29	s_310225_1602	3.2923	-3.5960	B	0.52	0.75	DES
30	s_61472_9399	359.5200	-54.6765	B	0.50	0.88	DES
31	s_137518_1133	31.3588	-35.6632	B	0.50	0.84	DES
32	s_353585_141	45.8222	+3.9613	B	0.50	0.84	—
33	s_43736_6056	67.3945	-60.1905	B	0.50	0.83	DES
34	s_257579_1725	23.3422	-12.8670	B	0.50	0.82	DES
35	s_289350_3716	164.8051	-7.1449	B	0.50	0.81	—
36	s_148704_6732	183.8176	-33.6006	B	0.50	0.80	—
37	s_301694_7426	26.2679	-4.9310	B	0.48	0.85	H20
38	s_58263_3751	42.7179	-55.4032	B	0.45	0.88	DES
39	s_194944_4494	59.2045	-24.1447	B	0.45	0.85	DES
40	s_280236_3353	25.8623	-8.8393	B	0.45	0.84	AGEL
41	s_90098_8002	107.1696	-46.7832	B	0.45	0.83	—
42	s_82090_1248	14.4037	-48.8040	B	0.45	0.81	DES
43	s_502450_1680	161.1145	+31.2344	B	0.45	0.79	—
44	s_426295_4804	182.0229	+16.6326	B	0.45	0.79	—
45	s_369309_653	30.4465	+6.6551	B	0.45	0.74	—
46	s_32666_26056	248.6090	-64.5569	B	0.45	0.73	—
47	s_156771_3446	54.3219	-31.8704	B	0.42	0.89	AGEL
48	s_94861_9433	21.9716	-45.5428	B	0.42	0.87	DES
49	s_102952_10164	334.8017	-43.8098	B	0.42	0.85	DES
50	s_90804_9850	3.9284	-46.6031	B	0.42	0.84	DES
51	s_71042_1653	20.1760	-51.7314	B	0.42	0.82	DES
52	s_266620_918	172.1320	-11.3108	B	0.42	0.80	—
53	s_242312_4967	50.3567	-15.5010	B	0.40	0.90	—
54	s_381188_1199	144.1511	+8.8633	B	0.40	0.90	H20
55	s_235296_3732	25.6457	-16.8049	B	0.40	0.88	AGEL
56	s_262449_2840	188.7861	-11.9798	B	0.40	0.87	—
57	s_91761_4679	350.3683	-46.5138	B	0.40	0.86	DES
58	s_360639_3636	13.4868	+5.3590	B	0.40	0.85	—
59	s_107312_5223	23.8451	-42.5398	B	0.40	0.85	DES
60	s_69087_268	307.2325	-52.5218	B	0.40	0.81	DES
61	s_165468_9648	64.7771	-29.9937	B	0.40	0.78	DES
62	s_256199_9097	29.6819	-13.0258	B	0.40	0.78	DES
63	s_364017_5199	141.0626	+5.7690	B	0.40	0.77	H20

continued...

Table 3 continued

#	name	RA	Dec	grade	$p_{\rm lens}$	v3blend8	known
64	s_188312_3322	33.2537	-25.3945	B	0.40	0.76	—
65	s_188754_2468	155.4695	-25.5029	B	0.40	0.74	—
66	s_119342_4136	50.7164	-39.7694	B	0.38	0.90	DES
67	s_405220_1292	136.4655	+13.0307	B	0.38	0.89	—
68	s_209576_5226	65.5760	-21.5461	B	0.38	0.86	DES
69	s_225679_7025	25.7203	-18.5211	B	0.38	0.79	AGEL
70	s_250611_2161	33.7898	-14.1185	B	0.38	0.77	—
71	s_161725_5181	61.4085	-30.6887	B	0.38	0.76	—
72	s_210799_616	33.4979	-21.2827	B	0.38	0.75	—
73	s_129372_3768	16.4504	-37.4286	B	0.35	0.82	DES
74	s_238185_2253	58.5761	-16.1645	B	0.35	0.82	AGEL
75	s_94933_7964	47.5548	-45.5743	B	0.35	0.78	DES
76	s_250997_4626	133.2708	-13.8944	B	0.35	0.77	—
77	s_71871_246	353.9664	-51.8716	B	0.35	0.76	AGEL
78	s_90920_9419	45.9517	-46.4407	B	0.30	0.85	DES

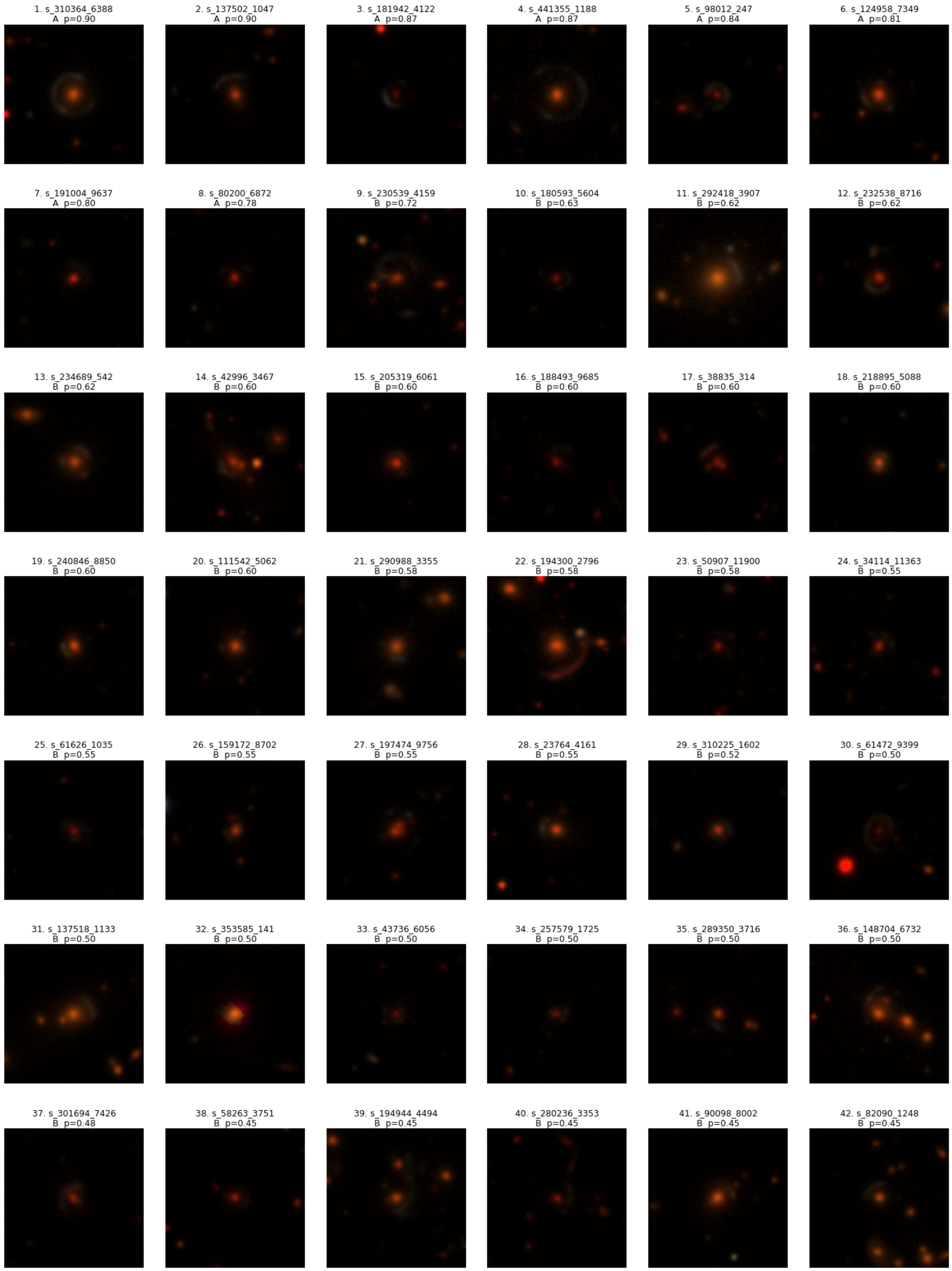


Figure 10: DESI *grz* cutouts ($0.262''$/px, Lupton RGB, 101 px) of the DESI-resolution A/B candidates, ranks 1–42 of Table 3. No high-resolution view exists for these positions.

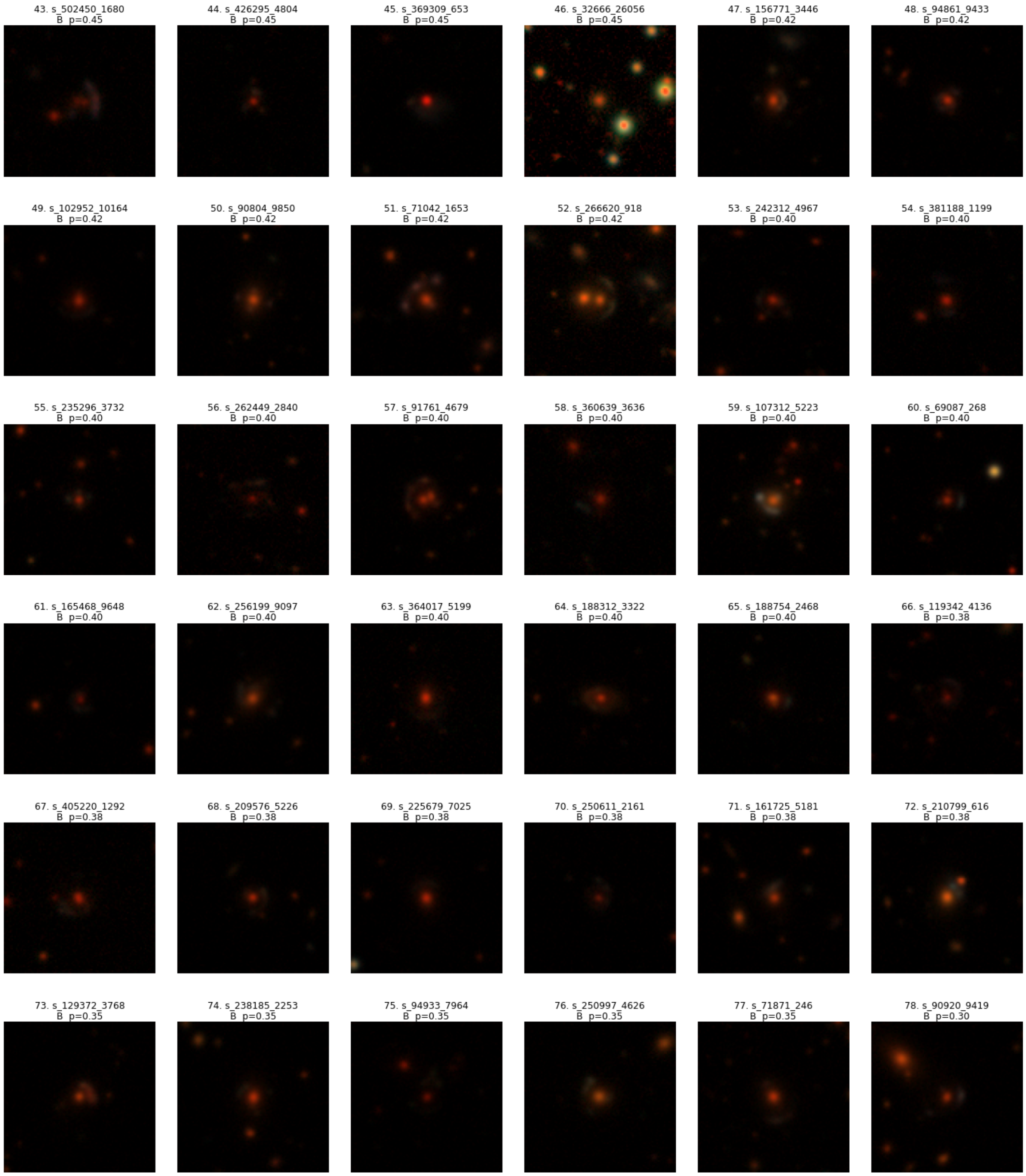


Figure 11: DESI *grz* cutouts (0.262″/px, Lupton RGB, 101 px) of the DESI-resolution A/B candidates, ranks 43–78 of Table 3. No high-resolution view exists for these positions.